

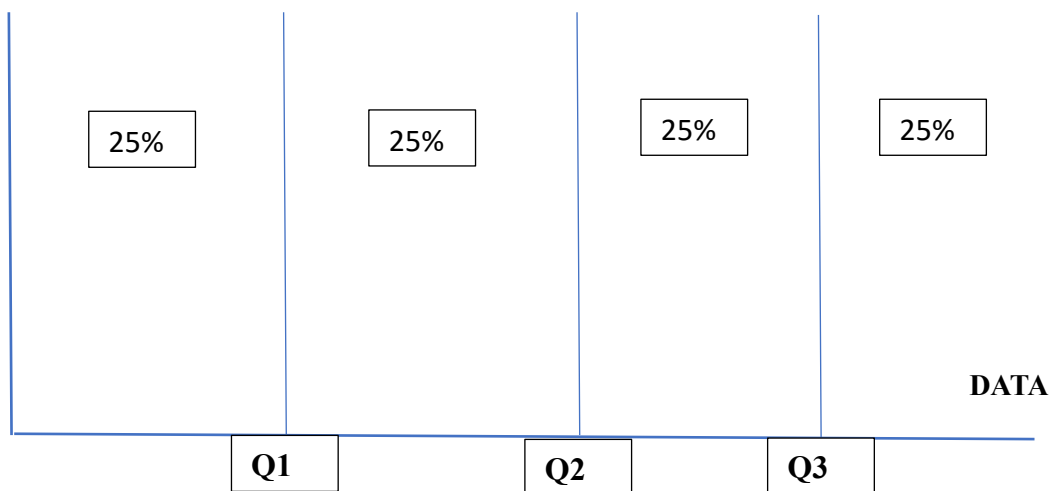
FIVE NUMBER SUMMARY USING PYTHON

- Descriptive analysis summarizes and organizes data to highlight key patterns using measures such as mean, standard deviation, and range.
- It provides a clear snapshot of the data's *central tendency, variability, and overall distribution* without making predictions or inferences.
- This analysis also helps us identify extreme data points or values that differ significantly from the rest of the dataset.
- Statistically, these are called *Outliers*.
- Such analysis can be visualized using a graphical representation called a *Box and Whisker Plot*, or presented numerically and called as "*Five Number Summary*".

Five Number summary has following components:

- Mean: - represent the average of the dataset.
- Min
- Lower Quartile (Q1): - 25% or 1st Quartile
- Median (Q2): - 50% or 2nd Quartile
- Upper Quartile(Q3): - 75%, 3rd Quartile.
- Max

Quartiles are specific percentiles that hold greater importance because they divide the data distribution into equal parts.



Some examples where this analysis is used include:

- Monitoring business trends.
- Understanding user engagement and content reach on social media.
- Summarizing patient health statistics.
- Analyzing student performance data.